

Stackwise: Rental Housing Law Navigator

Team Pioneer (solo entry: Ahmad Konainah). Hack-Nation x RealPage, Challenge 2. Not legal advice.

Live demo: <https://hack-nation-pioneer-seven.vercel.app/> · Code: <https://github.com/Ahmadkonainah/hack-nation-pioneer>

1. CHALLENGE TACKLED

Rental law is a stack of state statutes, city ordinances, pending bills and court rulings that change. User: a tenant, small landlord or housing counselor who needs to know **which rules apply at one address on one date**, with the source, and an honest "unknown" when the facts are missing. We built extraction (A), address lookup (B) and change tracking (C) for 3 states and 10 cities.

2. TOOLS AND MODELS

- **Claude Sonnet 5.5** (structured outputs): reads each page into schema-valid rule records; groups repeats and finds yields-to and conflict links; writes plain-English summaries and testable coverage conditions.
- **Python** (stdlib + Anthropic SDK): quote verification, date logic, three-valued lookup engine, tests. **JavaScript** port of the engine for the live demo.
- **U.S. Census geocoder**: state, county and legal city for each address. **Playwright**: browser checks of the demo.

3. WHAT WORKED WELL

- 56 rules from 67 pages, every quote verified in its source (500 addresses, 4,773 answers).
- All five change tests pass: T1 250, T2 90 (none in Newark), T3 140 with 90 flagged, T4 110 pending, T5 empty.
- Unknown is first-class: 14% of answers, each saying which fact is missing.
- Python and JavaScript engines agree on 7,000 of 7,000 date answers and 4,000 of 4,000 what-if answers. Over 40 self-checks and 72 unit tests pass in CI.
- A what-if panel lets a user type the year built or unit count and settle an "unknown" on the spot. Nothing typed leaves the browser.
- A date slider with Play runs the same engine across 2024 to 2028, so you can watch answers flip on the day a law starts.

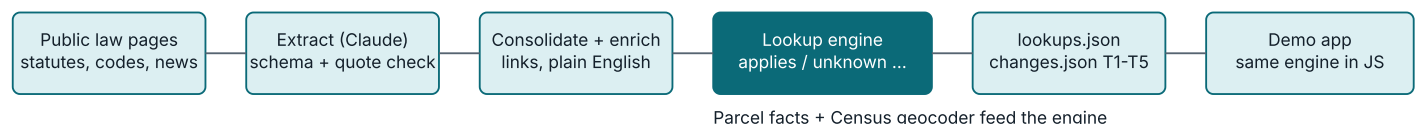
4. WHAT WAS CHALLENGING

- Parcel data lacks year built and units for many rows, and cannot show deed-restricted or owner-occupied status. Fix: three-valued logic and an explicit "not checked" list.
- Early model output over-linked laws (a state rule marked superseded by every city rule). Fix: tighter prompts, code repair, tests.
- Some code sites are not machine-readable. Fix: pages were read by hand and saved as text, never scraped. One saved page was a whole website (it would have cost \$6 alone), so the pipeline now skips pages that long.

5. HOW THE TIME WAS SPENT (IN ORDER)

Compared the challenges and chose this one → read the brief, data and tests → extraction with quote verification → consolidation, enrichment and geocoding → lookup engine and change tests → demo app and browser tests → README, videos and submission.

ARCHITECTURE



REFLECTION

A legal tool earns trust by what it refuses to guess. With 24 more hours: lawyer review of flagged conflicts, official code text for rules that rest on news summaries, and a native-speaker review of the Spanish plain-language view.